GLUCAGON

Severe hypoglycaemia

β-blocker overdose antidote

CHEMICAL NAME Recombinant human glucagon: 29-amino-acid polypeptide (His-Ser-Gln-Gly-Thr-Phe-Thr-Ser-Asp-Tyr-Ser-Lys-Tyr-Leu-Asp-Ser-Arg-Arg-Ala-Gln-Asp-Phe-Val-Gln-Trp-Leu-Met-Asn-Thr)

CLASS Pancreatic hormone (hyperglycaemic agent)

TRADE NAMES GlucaGen HypoKit · Baqsimi (nasal) · Gvoke · Glucagon Emergency Kit

THERAPEUTIC USES / INDICATIONS

- Severe hypoglycaemia in unconscious/uncooperative patient when IV access unavailable.
- β-blocker and calcium-channel blocker overdose (high-dose IV).
- Diagnostic aid: relax GI tract for radiology/endoscopy.
- Anaphylaxis unresponsive to adrenaline in patients on β-blockers.

ADMINISTRATION

- **IM/SC/IV**: reconstitute powder with supplied diluent, use immediately.
- **Nasal powder** (Baqsimi) into one nostril — no inhalation needed.
- Place unconscious patient in recovery position (vomiting).
- Give oral carbohydrate as soon as patient can swallow.

DOSAGES

- **Hypoglycaemia (adult & child >25 kg or ≥6–8 yr):** 1 mg SC/IM (or IV); may repeat once after 15 min (US label).
- **Child <25 kg or <6–8 yr:** 0.5 mg.
- **Nasal (Baqsimi):** 3 mg into one nostril (≥1 yr); may repeat after 15 min.
- **β-blocker overdose (adult):** 5 mg IV over 3–5 min, repeat once if needed (max 10 mg), then 5–10 mg/h infusion; child 0.05–0.15 mg/kg (max 5 mg).

PHARMACOKINETICS

- **Onset:** IV 1 min; IM/SC 8–10 min. **Duration:** 60–90 min.
- Degraded in liver, kidney and plasma. **t½:** 8–18 min.

PHARMACODYNAMICS

- Stimulates **hepatic glycogenolysis** and gluconeogenesis via glucagon receptors (cAMP) → ↑ blood glucose.
- Positive inotrope/chronotrope bypassing β-receptors; relaxes GI smooth muscle.
- Ineffective if glycogen stores are depleted (starvation, alcohol, adrenal insufficiency, chronic hypoglycaemia).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea and vomiting.
- Headache.
- Transient tachycardia.

ADVERSE EFFECTS (SERIOUS)

- Hypokalaemia (high doses).
- Hypersensitivity.
- Rebound hypoglycaemia (if no oral carbohydrate given).

CONTRAINDICATIONS & CAUTIONS

- Phaeochromocytoma (catecholamine release → hypertension).
- Insulinoma (paradoxical hypoglycaemia).
- Glucagonoma.

KEY INTERACTIONS

- **Warfarin:** ↑ anticoagulant effect (high doses).
- **β-blockers:** glucagon is used to reverse them.
- **Indomethacin:** may cause hypoglycaemia.

NURSING MANAGEMENT

- Check glucose; give glucagon if unconscious and no IV access.
- Place in recovery position; expect response in 10–15 min.
- Once awake: give oral fast-acting carbohydrate then a long-acting snack.
- If no response in 10–15 min: IV glucose, call for help.
- Investigate cause; sulfonylurea hypos need admission.

PATIENT EDUCATION

- Family should be trained to use the kit; check expiry date.
- After recovery, eat a snack containing starch.
- Nausea after the injection is common.
- Inform your diabetes team after any severe hypo.

KEY POINTS

- Raises glucose via glycogenolysis.
- Needs glycogen stores — fails in alcohol/starvation.
- 1 mg IM (0.5 mg small child); nasal 3 mg.
- Antidote for β-blocker overdose.

SOURCES CHECKED **[624]** Québec antidote guide **[625]** JBDS-IP hypoglycaemia guideline (2018) **[626]** Baqsimi, nasal (US label) **[627]** Glucagon, IM, IV (US label) **[628]** Gvoke VialDx, IV (US label)

Full references p. 432–457